

Encephalitis, myelitis, and acute disseminated encephalomyelitis (ADEM): Case definitions and guidelines for collection, analysis, and presentation of immunization safety data[☆]

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1. Preamble

1.1. Need for developing a standardized case definition and guidelines for encephalitis/acute disseminated encephalomyelitis as adverse events following immunization

Among the various events reported as adverse outcomes following immunizations, neurologic adverse events follow-

ing immunization (AEFI) are perhaps the most severe, and also perhaps the most difficult to address. The multifaceted presentation of neurologic illness, and relative lack of familiarity with the approach to and diagnosis of neurologic disease by many clinicians, makes neurologic AEFI some of the most challenging issues in clinical vaccinology. Further, the severity of central and peripheral nervous system events in individual patients often heightens the concern when such illnesses are associated with antecedent immunizations.

To improve comparability of vaccine safety data, the Brighton Collaboration *Encephalomyelitis/ADEM* Working Group has developed a case definition and guidelines for encephalitis, myelitis, and acute disseminated encephalomyelitis (ADEM), applicable in study settings with different availability of resources, in health care settings that differ by availability of and access to health care, and in different geographic regions.

The estimated incidence of *encephalitis*, in general, has varied widely, and is dependent upon age, demographics,

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season, causative agent, and presence of epidemic illness. Nearly all previous studies on encephalitis have been hospital-based. Estimates of viral encephalitis rates have ranged from 0.08/100,000 population in national passive surveillance studies [1] to 1–6 cases per 100,000 in hospital-based studies to 7.4 cases per 100,000 in one population-based study [2–4]. The incidence of encephalitis has generally been lower than that of aseptic meningitis, and children are at higher risk than adults. Case-fatality rates and incidence of severe neurologic sequelae also vary, dependent upon study methods and underlying etiologic agent. Population-based and hospital-based estimates of neurologic sequelae have ranged from 0.35 to 2.7 per 100,000 cases, with case fatality rates from 1 to 10% [2–5]. Certain agents are associated with higher case fatality rates; mortality rates of up to 33% have been reported with Herpes simplex encephalitis [6].

The underlying causes of acute encephalitis/myelitis are legion, and include infectious, toxic, neoplastic, autoimmune, and metabolic etiologies [7–13]. Most cases of encephalitis are thought to be infectious in nature, and may be attributed to a number of different viral, bacterial, fungal, and parasitic agents [7–10]. Toxic and metabolic agents may lead to a chemical encephalitis [10,11], while neoplasm may lead to either neoplastic or paraneoplastic encephalitis [6]. Finally, various autoimmune conditions may lead to acute encephalitis [10]. Immunizations may very rarely lead to acute encephalitis, particularly in the setting of live-attenuated viral vaccines [14–16]. However, the introduction of immunizations has served to reduce the incidence of encephalitic complications in several viral and bacterial infections [17–20]. In more than 70% of cases of acute encephalitis, however, a demonstrable etiologic agent is unable to be identified [13], presumably due to the broad range of possible underlying etiologies and limitations in current diagnostic testing.

Several different antecedent viral infections or vaccinations have been suggested as presenting an antigenic challenge leading to the immunologic response in the form of ADEM. A number of different viral infections, including measles, mumps, rubella, varicella-zoster, Epstein–Barr virus, cytomegalovirus, herpes simplex virus, Hepatitis A virus, and coxsackie virus have been associated with ADEM, with associated incidence rates varying from 1 per 1000 infections for measles and varicella to 1 per 5000 infections for rubella [21–23]. Additionally, a number of bacterial infections have been temporally associated with subsequent ADEM [24,25].

Various immunizations have been temporally associated with subsequent ADEM, including Japanese encephalitis, yellow fever, measles, influenza, smallpox, anthrax, and others [21,26,27]. However, the only epidemiologically and pathologically proven association of an antecedent event is with antirabies vaccination using the Semple rabies vaccine (a vaccine derived from sheep/mouse brains); such association has not been observed with modern formulations of rabies vaccine [28].

Although the precise pathophysiology leading to ADEM is unknown, a number of mechanisms have been proposed to explain the demyelination and immunogenic etiology of ADEM. Virus or viral products, or vaccine-associated products, could theoretically damage myelin membranes directly [29]. Infection of surrounding myelin-supporting cells by virus could lead to insertion of virus-specified polypeptides into host cell membranes, resulting in a humoral or cell-mediated autoimmune response to the infected cell; alternatively, infection of myelin-supporting cells could introduce sequestered myelin antigens into the circulation, inciting autoimmunity [30]. Molecular mimicry, a situation in which viral/vaccine polypeptides share common antigenic determinants with central or peripheral nerve myelin, could result in a distant autoimmune reaction [31]. Finally, infection/immunization may result in perturbation of immunoregulatory mechanisms, interfering with self-tolerance of host myelin proteins [32].

To date, no standardized case definition outlining the features of encephalitis/myelitis and ADEM has been developed for widespread use in surveillance, clinical trials, and diagnosis. Previously developed working case definitions for encephalitis used in various studies appearing in the literature have largely centered upon the demonstration of altered mental status in the presence of fever or other indicators of inflammation [7–13]. A definition published by Ball et al. [33] created for use in the U.S. Vaccine Injury Compensation program (<http://www.hrsa.gov/vaccinecompensation/>) was used in evaluating reports of suspected encephalitis submitted to the U.S. Vaccine Adverse Events Reporting System (VAERS), a passive vaccine safety surveillance system operated jointly by the U.S. Centers for Disease Control and Prevention (CDC) and the U.S. Food and Drug Administration (FDA) [34]. This definition was designed to categorize potential reports, and to identify key items of missing information. The case definitions proposed herein differ from previous case definitions primarily in the designed specificity of higher level (e.g., Level 1) criteria for encephalitis and ADEM. These higher levels of diagnostic certainty will most likely be of most utility in circumstances in which tertiary care and diagnostics are readily available; however, lower levels of diagnostic certainty have also been designed to achieve sufficient specificity even in the absence of resource-intensive diagnostic methods.

1.2. Methods for the development of the case definition and guidelines for encephalitis/ADEM as adverse events following immunization

Following the process described in the overview paper [35], a Brighton Collaboration *Encephalitis/ADEM* Working Group was formed in October 2003 with 14 members with public health, regulatory, clinical and academic, and industry backgrounds. The Working Group identified the key clinical and epidemiologic features required for case definitions for encephalitis, myelitis, and ADEM. Aspects of the necessary

criteria for classification of encephalitis/ADEM were based upon group discussion and consensus on each point; these decisions were, to the fullest extent possible, based upon evidence available from peer-reviewed literature and original research. The member composition and results of the web-based surveys completed by the reference group with subsequent discussions in the Working Group can be viewed at: http://www.brightoncollaboration.org/internet/en/index/working_groups.html.

To guide the decision-making for the case definition and guidelines, a literature search was performed by a Cochrane Collaboration professional search person for local reactions overall including swelling in the context of immunization (MEDLINE 1976–2004; search terms included amongst others “encephalitis”, “myelitis”, “encephalomyelitis/acute disseminated”, “ADEM”, “encephalopathy”, “meningoencephalitis”, “vaccine”, and “immunization”); the search also included reviews of textbooks and study protocols. Based on a review of the title and abstract of the >3000 references identified, we reviewed >220 potentially relevant articles. Case definitions for encephalomyelitis and ADEM that have previously appeared in the literature were reviewed in detail, and the salient aspects of each that were considered by the Working Group as central or core to any case definition for encephalitis/ADEM were incorporated.

1.3. Rationale for selected decisions about the case definitions of encephalitis and ADEM

Encephalitis is defined as inflammation of the parenchyma of the brain. Strictly speaking, it is a pathologic diagnosis, in which the presence of inflammation, edema, and neuronophagia (neuronal cell death) is demonstrated by histopathology. This is generally associated with an infiltration of inflammatory cells and perivascular cuffing. *Myelitis*, by corollary, is defined as inflammation of the parenchyma of the spinal cord. An inflammatory process involving both the brain and the spinal cord may be referred to as “*encephalomyelitis*”. Demonstration of this inflammation, regardless of the underlying cause, is considered confirmatory in the diagnosis of encephalitis/myelitis.

Previous definitions of encephalitis have been limited by rather non-specific descriptors of what constitutes encephalopathy or global cerebral dysfunction. While it is recognized by the Working Group that the term *encephalopathy* may have differences in connotation among various groups and in different contexts, for the purposes of these definitions, the term “encephalopathy” will be used to refer to a state of being, in which consciousness or mental status is altered, whereas “encephalitis” will refer to the specific neuropathologic state of cerebral parenchymal inflammation that is defined herein. In order to provide objective criteria in which global impairment of consciousness may reliably be documented by multiple observers, we have provided specific criteria indicative of such global impairment. These criteria should allow for specific documentation of evidence

of encephalopathy, and minimize inter-observer variability in interpretation of mental status. As seizures may result from cerebral inflammation and subsequent cortical irritation, and may often be present in patients with encephalitis [36], seizures associated with loss of consciousness have been included as a feature of global cerebral dysfunction.

As mentioned, encephalitis may be confirmed by demonstration of parenchymal brain inflammation by histopathology; such demonstration in effect provides a definitive diagnosis of encephalitis. However, it is recognized that in most cases, such histopathologic confirmation will be unavailable. The encephalitis Working Group recognizes that certain clinical and neurodiagnostic features may also be suggestive of acute encephalitis, allowing for a clinical diagnosis in the absence of tissue histopathology. It is this clinical diagnosis that is expected to be employed in the vast majority of situations utilizing these definitions.

A clinical diagnosis of encephalitis depends on the demonstration of brain dysfunction in the presence of evidence of an inflammatory process; this dysfunction may be cortical, subcortical (e.g., deep gray nuclei, brainstem), or both. Cerebral or cortical/subcortical dysfunction is generally manifested as (a) diffuse or global cortical dysfunction resulting in alteration of consciousness, impairment of mentation, or acute alteration of a person’s normal behavior, (b) focal cortical/subcortical dysfunction, resulting in so-called focal neurologic deficits, or (c) manifestations of both global dysfunction and focal deficits. Focal neurologic deficits are generally defined by dysfunction that can be localized to a specific location in the nervous system, resulting in neurologic abnormalities in a specific location, or in a specific function. Such deficits are generally referable to one of the major categories of neurologic activity—cortical, general motor, general sensory, autonomic, cerebellar, reflexes, or cranial nerves. There are many possible manifestations of focal neurologic deficits and altered cerebral function. Data on the specific neurologic signs and features seen in biopsy-confirmed cases of encephalitis are limited [37], but suggest that signs referable to nearly any aspect of the neuraxis may be observed in encephalitis/myelitis. Thus, the Working Group has attempted to outline a comprehensive list of various manifestations of dysfunction of the cortical, subcortical, motor, sensory, cerebellar, and cranial nerve systems in an encompassing manner in drafting the definition; however, the Working Group also recognizes that the relative sensitivities and/or specificities of these findings with respect to encephalitis are not known, and may not carry equal weight with respect to reflecting a diagnosis of encephalitis. In addition, it is possible that deficits not necessarily included in the list provided in the definition may also be present. Such an example may be focal, non-convulsive seizures.

The second aspect of clinical encephalitis is the demonstration of evidence of an acute inflammatory process. Fever is generally regarded as an overall indicator of inflammation; however, it is non-specific. The most reliable indicator of central nervous system (CNS) inflammation, in the

absence of histopathology, is demonstration of pleocytosis (presence of leukocytes in the cerebrospinal fluid [CSF]). In addition, there are various findings on neuroimaging (computed tomography [CT] or magnetic resonance imaging [MRI]) that are suggestive or indicative of parenchymal inflammation, and patterns and waveforms demonstrated on electroencephalography (EEG) that may be suggestive of an inflammatory CNS process. All of the above may apply to the diagnosis of myelitis, the main difference being that the clinical features of myelitis entail focal neurologic deficits, referable to the spinal cord; in myelitis alone, the cortex is not involved, and thus global cerebral dysfunction would not be expected. With respect to these indicators of inflammation, the Working Group has attempted to make the list of criteria succinct and as specific as possible. Fever (temperature $\geq 38^{\circ}\text{C}$ as defined by the Brighton Collaboration; http://www.brightoncollaboration.org/internet/en/index/definition_guidelines.html), while non-specific, is an easily measurable and reliable indicator of an inflammatory process; in the presence of the various additional criteria provided for nervous system dysfunction, it could allow for identification of likely cases of encephalitis without being overly specific. Limited data suggest that over 90% of biopsy-confirmed cases of herpes simplex encephalitis present with fever [37].

As mentioned, the most reliable indicator of CNS inflammation is pleocytosis. While not an absolute requirement for the diagnosis of encephalitis, lumbar puncture (LP) is often the most practicable method of assessing presence of CNS inflammation in most settings, including those with limited resources. In the absence of contraindications such as intracranial hypertension, LP and CSF analysis is the most helpful diagnostic test in cases of suspected encephalitis. A certain percentage of cases of encephalitis will not have an associated pleocytosis; the percentage of such cases would be expected to be small; however, the possibility of this occurrence should be recognized [38]. A cutoff of five leukocytes/ μL to define pleocytosis is used in many textbooks and is based on data on the normal distribution of leukocytes in healthy adults and children [39–42]. According to several studies [39–46], leukocyte reference values in infants younger than 2 months of age are considerably higher. A cutoff value of 15 leukocytes/ μL in children under 2 months of age was chosen based on several studies of healthy infants or infants without CNS infection [39–42]. Other studies suggesting even higher reference values did not use PCR or viral isolation to exclude viral etiology in infants with suspected CNS infection. By using these cut offs we estimated a false positive rate of 0–2% in adults [40–42] and 0–5% in neonates [39–42]. Higher cut off values would likely decrease sensitivity (thus increasing the risk of excluding cases of encephalitis) since in 5% of children with a diagnostically confirmed viral agent in CSF, a leukocyte count of less than $10\ \mu\text{L}^{-1}$ is present.

Traumatic LP occurs when the spinal needle used for the CSF draw penetrates the vascular epidural space and con-

taminates the CSF sample with blood. The visual threshold for blood contamination is approximately 400 red blood cells (RBC)/ μL , which is often used as a working definition for traumatic LP. Traumatic LP makes interpretation of white blood cell (WBC) count of a CSF sample difficult since peripheral WBCs from blood are likely to be introduced into CSF, making differentiation of the presence of WBCs due to intrathecal production or infiltration difficult. Several methods for interpretation of CSF parameters following traumatic LP have been suggested [47–51]. Predicted CSF WBC count can be estimated on the basis of peripheral blood count and CSF RBCs. True CSF pleocytosis is present if the predicted CSF WBC count exceeds the observed count. Alternatively, a simple leukocyte:erythrocyte ratio of 1:100 in CSF is highly sensitive and specific in diagnosing meningitis cases with more than 500 erythrocytes/ μL in CSF [52].

Evidence about increased leukocytes in CSF following seizures is conflicting but recent studies suggest that the underlying conditions responsible for the seizures are more likely to explain previously reported pleocytosis in these cases [53]. Therefore, the same cutoff value is used in this subgroup. The presence of polymorphonuclear leukocytes in CSF was not used as a criterion since data suggest that they can be frequently found in healthy adults on cytocentrifuged differential cell counts of CSF and often caused by even minimal blood contamination [54].

In addition to pleocytosis, neurodiagnostic criteria indicative of CNS inflammation have been included as indicators of encephalitis. These include demonstration of parenchymal inflammation on head CT or brain MRI scanning, and EEG patterns suggestive of diffuse or focal cortical dysfunction and inflammation. The specific criteria for the features of these studies that are “indicative of encephalitis” have been purposely left unspecified and open to interpretation, as the possible features indicative of encephalitis are multiple, and it is impractical to attempt to specify all findings that would be indicative of encephalitis. Personnel utilizing the definition are encouraged to consult with a neuroradiologist or neurologist when implementing these criteria.

Encephalitis is often associated with concomitant meningitis [1] since inflammation of meninges is often present along with encephalitis. However, from a clinical standpoint, the clinical features of meningitis (e.g., nuchal rigidity, phono-/photophobia, etc.) would in most cases be difficult or impossible to identify in a severely encephalopathic patient, and the clinical picture of encephalopathy will dominate. Diagnostically, parenchymal inflammation may frequently obscure the neuroimaging or electroencephalographic features of meningitis, making such separation difficult diagnostically. However, cases of encephalitis may often meet criteria for aseptic meningitis as well. While the majority of cases of aseptic meningitis are associated with favorable outcome and few sequelae [1], encephalitis may be associated with severe neurologic illness, serious long-term neurologic sequelae, and death. If there is evidence, clinically or diagnostically, of both meningeal and parenchymal

inflammation, and the case meets both encephalitis and aseptic meningitis [55] case definitions, it should be reported as encephalitis. The use of the term “*meningoencephalitis*” is frequently used in the literature and clinical practice, and is acceptable terminology in the case that both encephalitis and aseptic meningitis case definitions are met.

ADEM is classically described as a uniphasic syndrome of brain inflammation and demyelination, occurring in temporal association with an antecedent immunologic challenge, such as infection or an immunization. In general, ADEM is distinguished from acute encephalitis by (a) a predominance of demyelinating, rather than cytotoxic injury, and (b) a temporal association with a specific inciting immunogenic challenge [21,56,57].

The diagnostic hallmark of ADEM is the demonstration of scattered, focal or multifocal (disseminated) areas of inflammation and demyelination within cerebral subcortical and deep cortical white matter; while gray matter involvement may be seen as well (particularly in the thalamus), this is generally a minor component compared with white matter disease [58]. While these lesions may be demonstrable by histopathology, it is far more likely that MRI will be the diagnostic modality by which the diagnosis is made, and there are several MRI features that are helpful in distinguishing the white matter lesions of ADEM.

Most of the features and criteria as outlined above for encephalitis hold equally true for ADEM. In particular, it is important to note that the clinical manifestations of ADEM and acute encephalitis (global cerebral dysfunction, multifocal neurologic findings, meningismus) may be identical and difficult to distinguish on clinical grounds alone, particularly in the early phases of illness [21,56]. These two syndromes are also linked by the commonality of the presence of an encephalopathy or focal neurologic deficits and the presence of CNS inflammation. The key feature helping to distinguish these two from each other is the presence of acute demyelination in ADEM; such demyelination would primarily be documented by MRI findings. While demyelination may certainly be present in cases of acute viral encephalitis, the picture is generally predominated by associated edema, cytotoxic changes, and hemorrhage; MRI may also be normal.

ADEM may fall along a continuum of CNS demyelinating disorders that includes multiple sclerosis (MS), acute hemorrhagic leukoencephalitis, transverse myelitis, and optic neuritis. A certain percentage of persons with an initial diagnosis of ADEM will go on to evolve into MS [26], and as such, a comment on the monophasic nature of ADEM is necessary. ADEM differs from MS on the basis of absence of a relapsing or recurring pattern of clinical symptomatology and of new demyelinating lesions on MRI. It has been recognized, however, that in some cases of ADEM, the premature cessation or tapering of therapy (e.g., corticosteroids) may lead to a recurrence of symptoms. For this reason, the monophasic nature of ADEM will be defined as a lack of recurrence (within 3 months) *in the absence of treatment or while on appropriate treatment*. Relapse occurring during cessation or tapering of

therapy should be considered to belong to one monophasic episode. This is recognizing that, even with corticosteroid treatment, a subset of patients with suspected ADEM will nonetheless progress to MS. For cases of presumed ADEM, follow-up should continue for a period of >3 months, if possible, even in the setting of complete recovery from the initial neurologic episode.

Markers of CNS inflammation, however, (e.g., pleocytosis, neuroimaging features, EEG features) in ADEM are as has been stated for acute encephalitis, and will not be restated here. It should be noted that fever is frequently absent in cases of ADEM [59], and has not been listed as a criterion for evidence of inflammation for ADEM.

It is recognized by the Encephalitis/ADEM Working Group and should be emphasized to parents, patients, health care providers, and all others concerned with immunization safety, that encephalomyelitis or ADEM – or any other adverse event – which follows administration of an inactivated component or live vaccine may be temporally *associated with*, but is not necessarily *the result of*, administration of a vaccine. In particular, the presence of an antecedent, presumed initiating event has been an important aspect of many prior case definitions and classification schema for ADEM. However, because the definition itself defines a clinical entity without inference of a causal relation to a given exposure, the time interval from immunization until onset of the event cannot be part of the definition itself, but should be assessed as described in the guidelines. Accordingly, the time interval between immunization and onset of the event cannot be part of the definition itself, but should be assessed as described in the guidelines. As mentioned in the overview paper, the case definitions are structured in three different levels of diagnostic certainty. It should be noted that there are two paths each to reach Levels 2 and 3 of diagnostic certainty of encephalitis, and that Levels 2 and 3 only differ by number of indicators of inflammation of the CNS required for Level 2 compared to Level 3. Similarly, Levels 2 and 3 for myelitis only differ by the number of indicators suggestive of spinal cord inflammation required. There are also two paths to reach Level 1 of ADEM. Additionally, a Level 3 A has been created, if insufficient information is available to distinguish between encephalitis and ADEM. Generally, the clinical criteria case definitions have been drafted for use in all age groups. However, it is necessary to keep in mind that various neurologic and laboratory criteria used to define the presence of encephalopathy, as well as other manifestations of encephalitis, differ in infants and young children, in which the nervous system has not achieved the same level of development as in older children and adults [60,61]. By the age of 2 years, most of the aspects of the neurologic examination that are pertinent to a diagnosis of encephalitis have reached a point of maturation, and as such these neurologic examination features will not change significantly from 2 years onward.

The neurologic examination, CSF profile, and electroencephalographic and neuroimaging patterns continually

evolve after birth, most rapidly over the first weeks and months of life. The various features that define encephalopathy will differ depending on the specific age and development of the child. Level 1 of diagnostic certainty is a pathologic diagnosis, and may be achieved irrespective of age. However, the current case definitions for achieving Levels 2 and 3 of diagnostic certainty, which include clinical, electroencephalographic, and neuroimaging features defining encephalitis in adults and children 2 years of age and older, may differ significantly in infants and young children, and particularly in neonates. Depending on the post-gestational age and level of development of the child, some of the listed criteria may apply and others may not. For infants and children less than 2 years of age, the diagnosis of encephalopathy and encephalitis should be done using age-appropriate criteria. Whenever possible, evaluation by individuals who have specific neuropsychiatric or developmental and behavioral pediatric expertise will enhance the application and interpretation of these age-related criteria. As a result, the amount of confidence placed in an achievement of Levels 2 or 3 of diagnostic certainty in infants and children under 2 years may by necessity be less than that in adults and children 2 years and older. In order to assess the applicability of the case definition for encephalitis in children <2 years of age, we applied the case definition to 209 reports of encephalitis/encephalopathy in persons ≥ 2 years of age received by VAERS between 1991 and 2006, and to 244 reports in children <2 years of age, and found it to be equally applicable (data not published).

Also, while some of the clinical signs and symptoms listed in the definitions and guidelines may be subjective and culturally influenced, it should be recognized that this is an unavoidable part of standard medical practice.

The guidelines are structured according to the steps of conducting a study, i.e., data collection, analysis, and presentation. And similar to all Brighton Collaboration case definitions and guidelines, review of the definition with its guidelines is planned on a regular basis (i.e., every 3–5 years), or more often, if needed.

² If the lowest applicable level of diagnostic certainty of the definition for a definitive category (i.e., Level 3, excluding Level 3A) is met and there is evidence that the criteria of the next higher level of diagnostic certainty (Level 2) are met, the event should be classified in the next category. This approach should be continued until the highest level of diagnostic certainty for a given event can be determined. Thus, if a case fits diagnostic criteria for both categories (encephalitis and ADEM), but reaches a higher level of diagnostic certainty in one, the higher level supercedes, and the case should be classified according to the category in which the higher diagnostic certainty level is reached. The Working Group recognizes that under this paradigm, it is possible to reach a higher level of diagnostic certainty for ADEM with less stringent criteria than it is for encephalitis e.g., Level 1 diagnostic certainty for encephalitis requires histopathologic diagnosis, whilst ADEM Level 1 does not require this. However, in the absence of a biological marker, the diagnosis of ADEM rests upon the proper neuroimaging findings in the appropriate clinical context, and the combination of appropriate neuroimaging and a monophasic pattern of illness are as close to a gold standard as exist for this clinical entity. Thus, one may have a higher level of diagnostic certainty of ADEM than of encephalitis, in the absence of other biologic

2. Case definitions: encephalitis, myelitis, and ADEM²

2.1. Encephalitis

Level 1 of diagnostic certainty³: Encephalitis

- (a) Demonstration of acute inflammation of central nervous system parenchyma ($\pm$ meninges) by histopathology.

Level 2 of diagnostic certainty^{4,5}: Encephalitis

- (a) Encephalopathy (e.g. depressed or altered level of consciousness, lethargy, or personality change lasting >24 h),

AND INCLUDING

- (b) **ONE OR MORE** of the following:
 1. Decreased or absent response to environment, as defined by response to loud noise or painful stimuli,
 2. Decreased or absent eye contact,
 3. Inconsistent or absent response to external stimuli,
 4. Decreased arousability,
 5. Seizure associated with loss of consciousness [62].

OR

- (c) Focal or multifocal findings referable to the central nervous system, including one or more of the following:
 1. Focal cortical signs (including but not limited to: aphasia, alexia, agraphia, cortical blindness),
 2. Cranial nerve abnormality/abnormalities,⁶
 3. Visual field defect/defect(s),
 4. Presence of primitive reflexes (Babinski's sign, glabellar reflex, snout/sucking reflex),
 5. Motor weakness (either diffuse or focal; more often focal)⁶

data. When Level 1 ADEM and Level 2 encephalitis, or Level 2 ADEM and Level 3 encephalitis are met, the best category to choose would be ADEM.

³ The encephalitis/ADEM Working Group recognizes that, in most cases, histopathologic examination of tissue will not be practicable as a method of diagnosis; this may particularly be the case in developing countries. However, histopathologic demonstration of cerebral inflammation remains the "gold standard" for the diagnosis of encephalitis, and as such, the group has determined that this should be Level 1 for determination of encephalitis.

⁴ Levels 2 and 3 of diagnostics certainty have been especially designed for adults and children older than or equal to 2 years of age. For children under the age of 2 years (and, in particular, those under the age of 6 months) the nervous system and, as such, the neurologic examination is continually in flux (e.g., what is normal in a 28-day old is not necessarily normal in a 2-month old child). The evaluation of encephalopathy and neurologic deficits in infants and young children will need to be done in an age-appropriate fashion, taking into account the age and level of development of the child.

⁵ Levels 2 or 3 of encephalitis are met if criteria (a + b + d) or (c + d) from the respective levels are fulfilled, and no exclusion criteria are met.

⁶ Note that only criteria 2 and 5 may be applicable in all age groups; other criteria for focal/multifocal neurologic signs may be age-dependent, and will not be applicable to all age groups.

6. Sensory abnormalities (either positive or negative; sensory level),
7. Altered deep tendon reflexes (hypo- or hyperreflexia, reflex asymmetry),
8. Cerebellar dysfunction, including ataxia, dysmetria, cerebellar nystagmus.

AND (for both possibilities to reach Level 2)

- (d) **TWO OR MORE**⁷ of the following indicators of inflammation of the CNS:
1. Fever (temperature $\geq 38^{\circ}\text{C}$),
 2. CSF pleocytosis ($>5\text{ WBC/mm}^3$ in children >2 months of age; $>15\text{ WBC/mm}^3$ in children <2 months of age),
 3. EEG findings consistent with encephalitis,⁸ or
 4. Neuroimaging consistent with encephalitis.⁹

Level 3 of diagnostic certainty^{4,5}: Encephalitis

- (a) Encephalopathy (e.g. depressed or altered level of consciousness, lethargy, or personality change lasting $>24\text{ h}$),

AND INCLUDING

- (b) **ONE OR MORE** of the following:
1. Decreased or absent response to environment, as defined by response to loud noise or painful stimuli,
 2. Decreased or absent eye contact,
 3. Inconsistent or absent response to external stimuli,
 4. Decreased arousability, or
 5. Seizure associated with loss of consciousness [62].

OR

- (c) Focal or multifocal findings referable to the central nervous system, including **one or more** of the following:

1. Focal cortical signs (including but not limited to: aphasia, alexia, agraphia, cortical blindness),
2. Cranial nerve abnormality/abnormalities,⁶
3. Visual field defect/defect(s),
4. Presence of primitive reflexes (Babinski's sign, glabellar reflex, snout/sucking reflex),
5. Motor weakness (either diffuse or focal; more often focal),⁶
6. Sensory abnormalities (either positive or negative; sensory level),
7. Altered deep tendon reflexes (hypo- or hyperreflexia, reflex asymmetry), or
8. Cerebellar dysfunction, including ataxia, dysmetria, cerebellar nystagmus.

AND (for both possibilities to reach Level 3)

- (d) **ONE**⁷ of the following indicators of inflammation of CNS:
1. Fever (temperature $\geq 38^{\circ}\text{C}$),
 2. CSF pleocytosis ($>5\text{ WBC/mm}^3$ in children >2 months of age; $>15\text{ WBC/mm}^3$ in children <2 months of age),
 3. EEG findings consistent with encephalitis,⁸ or
 4. Neuroimaging consistent with encephalitis.⁹

Level 3A of diagnostic certainty^{6,10}

- (a) Insufficient information is available to distinguish case between acute encephalitis or ADEM; case unable to be definitively classified.

Exclusion criterion for levels 2 and 3 of diagnostic certainty

- (a) Other diagnosis for illness present.¹¹

2.2. Myelitis

Level 1 of diagnostic certainty: Myelitis

- (a) Demonstration of acute spinal cord inflammation ($\pm$ meninges) by histopathology.

Level 2 of diagnostic certainty¹²: Myelitis

- (a) Myelopathy (development of sensory, motor, or autonomic dysfunction attributable to the spinal cord,

⁷ Note that Level 2 of diagnostic certainty requires at least 2 of the listed criteria for inflammation, while Level 3 required only 1 criteria. This is in recognition that, in some cases of encephalitis, all listed criteria will either not be present, or such data will be unavailable. Thus, a clinical diagnosis of encephalitis should still be applicable, but will be of less diagnostic certainty than if sufficient criteria were present.

⁸ Electroencephalographic (EEG) findings consistent with encephalitis: EEG findings consistent with encephalitis include, but are not limited to: Diffuse or multifocal nonspecific (nonphysiologic) background slowing; periodic discharges or other encephalographic abnormalities may or may not be present.

⁹ Neuroimaging findings consistent with encephalitis: Neuroimaging findings consistent with encephalitis include, but are not limited to: head computed tomography (CT) displaying areas of hypodensity; contrast images demonstrating meningeal and parenchymal enhancement indicating meningeal and parenchymal inflammation, or gyral enhancement, brain/spine magnetic resonance imaging (MRI) displaying diffuse or multifocal areas of hyperintense signal on T2-weighted, diffusion-weighted image, or fluid-attenuation inversion recovery (FLAIR) sequences, suggestive of inflammation or demyelination.

¹⁰ In certain situations, insufficient information will be available to make a definitive distinction between acute encephalitis and ADEM; in such instances, the Level of diagnostic certainty 3A should be used, and all attempts should be made to obtain additional information that will allow for further categorization of the case.

¹¹ For example, neoplasm, toxic/metabolic encephalopathy, vascular disorder, trauma, etc.

¹² Levels 2 or 3 of myelitis are met if criteria (a + b) from the respective levels are fulfilled, and no exclusion criteria are met.

including upper- and/or lower-motor neuron weakness, sensory level, bowel and/or bladder dysfunction, erectile dysfunction).

AND

- (b) **TWO OR MORE**¹³ of the following indicators suggestive of spinal cord inflammation:
1. Fever (temperature $\geq 38^{\circ}\text{C}$),
 2. CSF pleocytosis ($>5\text{ WBC/mm}^3$ in children >2 months of age; $>15\text{ WBC/mm}^3$ in children <2 months of age),
 3. Neuroimaging findings demonstrating acute inflammation ($\pm$ meninges), or demyelination of the spinal cord.

Level 3 of diagnostic certainty¹²: Myelitis

- (a) Myelopathy (development of sensory, motor, or autonomic dysfunction attributable to the spinal cord, including upper- and/or lower-motor neuron weakness, sensory level, bowel and/or bladder dysfunction, erectile dysfunction).

AND

- (b) **ONE**¹³ of the following indicators suggestive of spinal cord inflammation:
1. Fever (temperature $\geq 38^{\circ}\text{C}$),
 2. CSF pleocytosis ($>5\text{ WBC/mm}^3$ in children >2 months of age; $>15\text{ WBC/mm}^3$ in children <2 months of age),
 3. Neuroimaging findings demonstrating acute inflammation ($\pm$ meninges), or demyelination of the spinal cord.

Exclusion criterion for Levels 2 and 3 of diagnostic certainty

- (a) Other diagnosis for illness present¹¹.

Cases fulfilling the criteria for both encephalitis and myelitis in any category would be classified as *encephalomyelitis*.

2.3. ADEM

Level 1 of diagnostic certainty¹⁴: ADEM

- (a) Demonstration of diffuse or multifocal areas of demyelination by histopathology.

¹³ Note that Level 2 of diagnostic certainty requires at least 2 of the listed criteria for inflammation, while Level 3 required only 1 criteria. This is in recognition that, in some cases of myelitis, all listed criteria will either not be present, or such data will be unavailable. Thus, a clinical diagnosis of encephalitis should still be applicable, but will be of less diagnostic certainty than if sufficient criteria were present.

¹⁴ Level 1 of ADEM is met if criterion (a) is met or if criteria (b + c + d) are fulfilled, and no exclusion criteria are met.

OR

- (b) Focal or multifocal findings referable to the central nervous system, including **one or more** of the following:
1. Encephalopathy (see case definition for encephalitis for specification of encephalopathy),
 2. Focal cortical signs (including but not limited to: aphasia, alexia, agraphia, cortical blindness),
 3. Cranial nerve abnormality/abnormalities,
 4. Visual field defect/defects,
 5. Presence of primitive reflexes (Babinski's sign, glabellar reflex, snout/sucking reflex),
 6. Motor weakness (either diffuse or focal; more often focal),
 7. Sensory abnormalities (either positive or negative; sensory level),
 8. Altered deep tendon reflexes (hypo- or hyperreflexia, asymmetry of reflexes), or
 9. Cerebellar dysfunction, including ataxia, dysmetria, cerebellar nystagmus,

AND

- (c) Magnetic resonance imaging (MRI) findings displaying diffuse or multifocal white matter lesions on T2-weighted, diffusion-weighted (DWI), or fluid-attenuated inversion recovery (FLAIR) sequences ($\pm$ gadolinium enhancement on T1 sequences),

AND

- (d) Monophasic pattern to illness (i.e., absence of relapse within a *minimum* of 3 months of symptomatic nadir^{15,16,17}).

¹⁵ Absence of recurrence will be fulfilled only through long-term follow-up assessment. Lack of recurrence of neurologic symptoms within a 3-month period of first episode would be considered suggestive of a monophasic illness. A certain percentage of cases, however, will likely later be classified as multiple sclerosis. Diagnostic certainty is increased in direct correlation with duration of follow up with absence of recurrence of illness.

¹⁶ Symptomatic nadir is defined at the point at which clinical symptoms are felt to be at the clinical worst; this nadir will need to be defined and identified by the health practitioner on a case-by-case basis; interval between illness onset and symptomatic nadir should be documented.

¹⁷ The absence of recurrence and monophasic nature of ADEM is useful as a distinguishing feature to discern ADEM from multiple sclerosis. It is recognized, however, that some authorities recognize an entity of "relapsing ADEM", which may represent a separate pathophysiologic entity from MS. However the Working Group decided that recurrence of illness following a 3-month interval would be more likely representative of MS, and for the operational purposes of this definition, such recurrence would be operationally considered MS, and thus "not a case" of ADEM. Additionally, the development of acute demyelinating illness in a person with a known history of MS may be difficult to classify as either ADEM or an exacerbation of MS; however, most authorities would likely classify such an event as an MS exacerbation, with the understanding that such differentiation may be imperfect and uncertain.

Level 2 of diagnostic certainty^{4,18}: ADEM

- (a) Focal or multifocal findings referable to the central nervous system, including one or more of the following:
10. Encephalopathy (see case definition for encephalitis for specification of encephalopathy),
 11. Focal cortical signs (including but not limited to: aphasia, alexia, agraphia, cortical blindness),
 12. Cranial nerve abnormality/abnormalities,
 13. Visual field defect/defects,
 14. Presence of primitive reflexes (Babinski's sign, glabellar reflex, snout/sucking reflex),
 15. Motor weakness (either diffuse or focal; more often focal),
 16. Sensory abnormalities (either positive or negative; sensory level),
 17. Altered deep tendon reflexes (hypo- or hyperreflexia, asymmetry of reflexes), or
 18. Cerebellar dysfunction, including ataxia, dysmetria, cerebellar nystagmus,

AND

- (b) Magnetic resonance imaging (MRI) findings displaying diffuse or multifocal white matter lesions on T2-weighted, diffusion-weighted (DWI), or fluid-attenuated inversion recovery (FLAIR) sequences ($\pm$ gadolinium enhancement on T1 sequences),

AND

- (c) Insufficient follow up time achieved to document absence of relapse within a minimum period of 3 months following symptomatic nadir.^{15,16,17}

Level 3 of diagnostic certainty⁴: ADEM

- (a) Focal or multifocal findings referable to the central nervous system, including one or more of the following:
19. Encephalopathy (see case definition for encephalitis for specification of encephalopathy),
 20. Focal cortical signs (including but not limited to: aphasia, alexia, agraphia, cortical blindness),
 21. Cranial nerve abnormality/abnormalities,
 22. Visual field defect/defects,
 23. Presence of primitive reflexes (Babinski's sign, glabellar reflex, snout/sucking reflex),
 24. Motor weakness (either diffuse or focal; more often focal),
 25. Sensory abnormalities (either positive or negative; sensory level),
 26. Altered deep tendon reflexes (hypo- or hyperreflexia, asymmetry of reflexes), or
 27. Cerebellar dysfunction, including ataxia, dysmetria, cerebellar nystagmus,

Level 3A^{6,10}

- Insufficient information is available to distinguish case between acute encephalitis or ADEM; case unable to be definitively classified.

Exclusion criteria for all levels of diagnostic certainty

- Presence of a clear alternative acute infectious or other diagnosis for illness,
- Recurrence or relapse of illness at any point following a 3 month period of clinical improvement from symptomatic nadir, or
- If known, MRI findings or histopathologic data inconsistent with the diagnosis of ADEM.

3. Guidelines for data collection, analysis, and presentation of encephalitis/acute disseminated encephalomyelitis as an adverse event following immunization

It was the consensus of the Encephalitis/ADEM working group to recommend the following guidelines to enable meaningful and standardized collection, analysis, and presentation of information about encephalomyelitis/ADEM.¹⁹ However, the group recognized that implementation of all guidelines might not be possible in all settings. The availability of information may vary depending upon resources, geographic region, and whether the source of information is a prospectively designed clinical trial, a post-marketing surveillance or epidemiologic study, or an individual case report. Also, as explained in more detail in the overview paper [35], these guidelines have been developed by this Working Group for guidance only, and are not considered a mandatory requirement for data collection, analysis, or presentation.

3.1. Data Collection

These guidelines represent a desirable standard for collection of data on encephalitis/ADEM to allow for comparability of data, and are recommended as an addition to data collected for the specific study question and setting. The guidelines are not intended to guide the primary reporting of encephalitis/ADEM to a surveillance system or study monitor. Investigators developing a data collection tool based on these data collection guidelines also need to refer to the criteria in the case definition, which are not repeated in these guidelines.

Guidelines 2,4,10,11,15,17–21 below have been developed to address data elements for the collection of adverse

¹⁸ Level 2 of ADEM is met if criteria (a + b + c) are fulfilled and no exclusion criteria are met.

¹⁹ Unless otherwise specified in the text, the term encephalitis will refer to both acute encephalitis and/or myelitis, and ADEM will refer to cases of ADEM with or without transverse myelitis.

event information as specified in general drug safety guidelines by the International Conference on Harmonization of Technical Requirements for Registration of Pharmaceuticals for Human Use (ICH) and the form for reporting of drug adverse events by the Council for International Organizations of Medical Sciences (CIOMS) [63,64]. These data elements include an identifiable reporter and patient, one or more prior immunizations, and a detailed description of the AEFI, in this case, of encephalitis/ADEM. The additional guidelines have been developed as guidance for the collection of additional information to allow for a more comprehensive understanding of these entities as AEFI.

3.1.1. Source of information/reporter

For all cases and/or all study participants, as appropriate, the following information should be recorded:

- (1) Date of report.
- (2) Name and contact information of person reporting, diagnosing, and/or observing the event in accordance with country-specific data protection laws.
- (3) Relation to the patient (e.g., immunizer [clinician, nurse], family member [indicate relationship], other).

3.1.2. Vaccinee/control

For all cases and/or all study participants, as appropriate, the following information should be recorded.

3.1.2.1. Demographics.

- (4) Case/study participant identifiers (e.g., first name initial followed by last name initial), or code (or otherwise specified by country-specific data protection laws).
- (5) Date of birth, age, sex, race/ethnicity (as appropriate).
- (6) For infants: gestational age and birth weight.

3.1.2.2. Clinical and immunization history.

- (7) The presence of any pre-immunization condition (in particular, pre-existing neurologic illness) that may affect the evaluation of encephalitis as an AEFI.
- (8) Any medication administered prior to, during, or after immunization by any route, biologics and prescription and non-prescription medication or treatment with long half-life (e.g., immunoglobulins and blood transfusion and specially given to treat the AEFI), including indication which ones were given to treat the event. In addition, the specific timing of the administration of the medication in relationship to the adverse event should be recorded.
- (9) Immunization history (i.e., previous immunizations and any adverse events following immunization ever experienced), in particular occurrence of similar event prior to immunization or after a previous immunization.

3.1.3. Details of the immunization

For all cases and/or all study participants, as appropriate, the following information should be recorded:

- (10) Date and time of immunization.
- (11) Description of vaccine(s) (name of vaccine, manufacturer, lot number, dose, volume [e.g., 0.25, 0.5 mL, etc.] and number of doses).
- (12) Geographic location (e.g., country and city/state, if applicable) in which vaccination was given.
- (13) Route and method of administration (e.g., oral, intramuscular, intradermal, subcutaneous, and needle-free or other injection devices), including size/gauge and type of administrative instrument.
- (14) If injected, the anatomical site(s) including left or right and exact location of each injection (e.g., vaccine A in proximal left lateral thigh, vaccine B in left deltoid).

3.1.4. The adverse event

For all cases and/or all study participants, as appropriate, the following information should be recorded:

- (15) For all cases at any level of diagnostic certainty and for reported events with insufficient evidence, the criteria fulfilled to meet the case definition and other signs and symptoms indicative of encephalitis/ADEM.

Specifically, a description of clinical manifestations and course, making careful note of the following:

- Which of the neuroimaging or histopathologic features, and/or the specific clinical findings referable to the central nervous system contributed to the classification of the case of encephalitis/ADEM.
- If multiple neurologic events occur at separate times, such as altered mental status, seizures, ataxia, etc., the initial neurologic findings leading to the first fulfilment of case definition criteria for encephalitis/ADEM, and subsequent neurologic signs recorded, ideally with their respective start and ending dates.
- Nature of onset of encephalopathy (acute [e.g., evolving over minutes–hours to hours–days], subacute [e.g., evolving over hours–days to days–weeks]).
- Level of consciousness at clinical nadir—this may be recorded using the Glasgow Coma Scale [Age-appropriate Pediatric Coma Score in children <5 years] (see [Appendix A.1](#)).
- Specific description of additional neurologic signs (e.g., ataxia, seizures, movement disorders).
- Results of all neuroimaging studies (e.g., computed tomography [CT] or magnetic resonance imaging [MRI]), including dates of performance [There are a number of MRI features that may be helpful in distinguishing the white matter lesions in ADEM—see [Appendix A.2](#)].
- Results of additional electrodiagnostic studies (e.g., electroencephalography [EEG], electromyogra-

- phy/nerve conduction velocity studies [EMG/NCS]), including dates of performance.
- Results of cerebrospinal fluid (CSF) examination, including white cell count (WBC, in cells/mm³), red cell count (RBC, in cells/mm³), differential leukocyte count (when available), protein level (in mg/dL), glucose level (in mg/dL), concomitant serum glucose level (in mg/dL), and including dates of performance.
 - Results of any additional laboratory testing that is performed which may be useful in identifying an etiology of encephalopathy/myelopathy other than encephalitis/ADEM (see [Appendix A.6](#)).
- (16) Concurrent signs, symptoms, and diseases other than the event described.
- (17) Date and time of
- Onset;²⁰
 - First observation²⁰;
 - Diagnosis²¹;
 - First prodromal symptom, and date of first neurologic symptom; and
 - All CT, MRI, EEG, EMG/NCS, CSF studies.
- (18) Methods of Measurement (as applicable in respective study setting):
- A. Neurologic consultation should be obtained when possible. Detailed notes from the neurologic examination, in addition to the scores listed in 19B below.
- B. Neurologic status using all of the following applicable parameters:
- Glasgow Coma Scale/Pediatric Coma Score (see [Appendix A.1](#));
 - Mini-Mental State Examination (when applicable) (see [Appendix A.3](#));
 - Barthel Index (see [Appendix A.4](#));
 - Modified Rankin Functional Score (see [Appendix A.5](#)); and
 - Notes from neurologic examination (obtained from neurologic consultation)
- (19) Frequency of measurement:
- A. Neurologic findings at the following intervals:
- Initial presentation to medical care;
 - At clinical nadir²²;
 - At all subsequent points of significant change in neurologic status until the end of the clinical course (recovery, death, or end of follow up), if possible, but at least at weekly intervals; and
 - Additional measurements will be determined by clinical course.
- B. Results of validated neurologic indices in the following fashion:
- At initial presentation to medical care;
 - Daily during the first week of hospitalization, or duration of hospitalization if shorter than 1 week;
 - Every 3 days following the first week of hospitalization if still hospitalized;
 - At discharge from acute care; and
 - At monthly intervals for 3–6 months following discharge from acute care (either if returned to home or if in assisted care facility).
- (20) Duration of monitoring: final outcomes in patients with encephalitis or ADEM will vary; duration of monitoring will depend on several factors, including apparent short- and long-term morbidity and mortality, severity of initial neurologic illness, and the practicality of long-term monitoring.

3.1.4.1. Encephalitis. The ultimate outcome in patients developing encephalitis may range widely, from complete recovery to persistent coma and death. A proportion of persons developing encephalitis will be expected to have persistent neurologic, functional, and cognitive sequelae lasting for months, years, or indefinitely. Ideally, patients should be followed until death (if acute), or until full recovery is achieved. In persons with persisting deficits, long-term monitoring may not be practicable or possible, and should be monitored as long as possible, with documentation of the duration of surveillance if lost to follow-up.

3.1.4.2. ADEM. As ADEM is defined as a monophasic illness, the likelihood of a monophasic nature to the illness is directly correlated with duration of monitoring. In order to meet Level 1 diagnostic criteria for ADEM, monitoring must be completed for a minimum of 3 months following clinical nadir; however, if possible, monitoring for a longer duration should be performed. Cases in which monitoring for at least 3 months is not possible may only be classified at Level 2 or Level 3.

(21) The outcome of encephalitis/ADEM as well as the timing relative to immunization and the time course of the evolution of the neurologic syndrome (including date of final outcome or final observation). The following details regarding outcome, using the following suggested terms:

A. Neurologic/Functional Outcome

- Recovered, no sequelae, back at premorbid baseline status;
- Recovered, neurologic sequelae present at time of final follow up;

²⁰ The date and/or time of onset is defined as the time post-immunization, when signs or symptoms suggestive of neurologic illness as self-reported by the patient or surrogate, or by documentation or observation by a health care provider, were first experienced and/or described. Date/time of onset is by necessity less valid than date of diagnosis.

²¹ The date of diagnosis of an episode is the day the event met the case definition as determined by study personnel or a healthcare professional.

²² Clinical nadir is operationally defined at the point at which clinical symptoms are felt to be at the clinical worst; this nadir will need to be defined and identified by the health practitioner on a case-by-case basis; interval between illness onset and symptomatic nadir should be documented.

- Died;
- Outcome unknown; or
- Other outcome (describe).

B. Disposition

- Disposition to home, independent living;
- Disposition to home, dependent living;
- Disposition to pre-illness residence other than home (nursing home, skilled facility, etc.), independent living or pre-illness baseline status;
- Disposition to assisted living or rehabilitation;
- Died;
- Disposition unknown; or
- Other disposition (describe).

C. Treatment

- Treatment rendered (antivirals, corticosteroids, intravenous immune globulin, plasmapheresis, etc.); or
- No treatment rendered.

3.1.5. Miscellaneous

- (22) The duration of surveillance for encephalitis, when collected as a pre-specified adverse event in clinical trials, is to some extent arbitrary and depends on
 - Biologic characteristics of the vaccine (e.g., live attenuated versus inactivated
 - component vaccines);
 - Biologic characteristics of the vaccine-targeted disease;
 - Biologic characteristics of both acute encephalitis and ADEM, including patterns
 - identified in previous trials (e.g., early phase trials); and
 - Biologic characteristics of the vaccinee.
- (23) Methods of data collection, including use of validated neurologic indices and documentation of neurologic examination, should be as consistent within and between study groups as is possible.
- (24) Follow up of cases should aim to verify and complete the information as outlined in guidelines 1–21.

3.2. Data analysis

These guidelines represent a desirable standard for the analysis of data on post-immunization encephalitis/ADEM to allow for comparability of data, and are recommended as an addition to data analyzed for the specific study question and setting.

- (25) Reported events should be classified in one of the following five categories. Events that meet the case definition should be classified according to the three levels of diagnostic certainty as specified in the case definition. Events that do not meet the case definition should be classified in the additional categories for analysis. It should be attempted to reach the highest level possible,

e.g., if Level 2 is fulfilled proceed to see if Level 1 is also fulfilled.

Event classification in five categories

Event meets case definition

Main categories with subcategory

- (1) Level 1: criteria as specified in the case definition for encephalitis/myelitis, or ADEM.
- (2) Level 2: criteria as specified in the case definition for encephalitis/myelitis, or ADEM.
- (3) Level 3: criteria as specified in the case definition for encephalitis/myelitis, or ADEM.
 - Level 3A: insufficient data provided to allow for a distinction between Level 3 encephalitis/myelitis and Level 3 ADEM.

Event does not meet case definition

Additional categories for analysis

- (4) Reported event of encephalitis/ADEM, with insufficient evidence to meet the case definition.^{23,24}
- (5) Not a case of encephalitis/ADEM.²⁵
 - To determine the appropriate category, the user should first establish whether a reported event meets the criteria for the lowest applicable level of diagnostic certainty (i.e., Level 3A). If sufficient data are provided to allow for a distinction between Level 3 encephalitis/myelitis and Level 3 ADEM, the category of distinction should be assessed, and if there is evidence that the criteria of the next higher level (Level 2) of diagnostic certainty are met, the event should be classified in the next category. This approach should be continued until the highest level of diagnostic certainty for a given event can be determined. If the lowest level of the case definition(s) is not met, it should be determined that any of the higher levels of diagnostic certainty are similarly not met, and the event should be classified in category 4 or 5.
 - In the scenario in which an event meets diagnostic criteria (above Level 3A) for both encephalitis/myelitis and ADEM, but reaches a higher level of diagnostic certainty for one or the other, the higher level of diagnostic certainty supercedes, and the event should be categorized according to this higher level of diagnostic certainty.

²³ If the evidence available for an event is insufficient to permit classification at any level of diagnostic certainty (e.g., because of missing information), such an event should be categorized as “reported encephalitis/ADEM with insufficient evidence to meet the case definition”. Notations should be made as to what evidence is missing.

²⁴ An event does not meet the case definition if investigation reveals a negative finding of a necessary criterion or if an exclusion criterion is met. Such an event should be rejected and classified as “Not a case of encephalitis/ADEM”.

²⁵ For example, total of 10 cases of encephalitis/ADEM in 2000 study participants.

- If there is evidence, clinically or diagnostically, of both meningeal and parenchymal inflammation, and the case meets both encephalitis and aseptic meningitis [55] case definitions, it should be classified as encephalitis.
- (26) The interval between immunization(s) and onset of encephalitis/ADEM should be determined from the date of immunization(s) and date/time of onset,²⁰ or first observation,²⁰ or diagnosis,²¹ whichever is available. Whatever dates are used, they should be used consistently within and across study groups. For limited numbers of cases, the exact time course should be assessed for each case. For large case series or data sets, total number of persons developing encephalitis/ADEM should be reported over the total number of subjects. All intervals should be analyzed and categorized by day (e.g., numerator/denominator [%] on day X post-immunization).
- (27) Individual case summaries or case reports represent the ideal method of assessment of each case of encephalitis/ADEM for few cases.
- (28) The following time intervals should be compared among patients (using mean, median, and range):
- Interval (days) between immunization(s) and onset of prodromal symptoms;
 - Interval (days) between immunization(s) and onset of neurologic signs;
 - Interval (days) between onset of neurologic signs and clinical nadir (as determined by the investigator);
 - Number of days with a Glasgow Coma Scale score less than 10; and
 - Interval (days) between onset of neurologic signs and collection of CSF (recording all such intervals if multiple CSF specimens collected).
- (29) The most critical items to analyze in case reports of encephalitis/ADEM include, but are not limited to, the following:
- The interval between the date of onset,²⁰ if known, or date of first observation²⁰ or date of diagnosis²¹ to the date of outcome at last follow up (mean, median, range);
 - Neurologic index scores (Appendix A) at the designated intervals, including: (mean, median, range)
 - At presentation;
 - At clinical nadir;
 - At day 1 through day 7, or for duration of hospitalization if less than 1 week;
 - At time of discharge from acute care, noting the number of days from admission to discharge from acute care;
 - At days 10, 13, 16, and every 3 days thereafter following first week of hospitalization, until acute care discharge; and
 - At each month for months 1–6 after acute care discharge.
 - The mean, median, and range for the following parameters of CSF profile should be analyzed for:
 - WBC (mm^3);
 - RBC (μL^{-1});
 - Protein (mg/dL); and
 - Glucose (mg/dL).
- (30) The following data could be compared, as best as possible, between patients:
- Number of patients meeting the case definitions for encephalitis/ADEM with the following additional neurologic signs at any point in illness
 - Coma;
 - Seizures;
 - Focal weakness;
 - Tremors;
 - Myoclonus;
 - Ataxia;
 - Dysarthria/dysphagia; or
 - Cranial nerve abnormalities.
 - Number of patients with
 - Abnormal CT findings;
 - Abnormal MRI findings;
 - Abnormal EEG findings;
 - Abnormal EMG/NCS findings; or
 - Abnormal CSF profiles.
 - Number of patients
 - Receiving antiepileptic medication;
 - Receiving corticosteroids;
 - Receiving intravenous immune globulin;
 - Receiving plasmapheresis; or
 - Receiving antimicrobial agents/antiviral medication.
 - Clinical and functional outcome
 - Number of patients who died, who were discharged to home, who were discharged to long-term care facilities [dispositions outlined in (21B)]
 - Number of patients who survived (with and without neurologic sequelae), who died, and who had other outcomes [functional outcomes outlined in (21A)]
- (31) Data on encephalitis/ADEM should be compared with those obtained from one or more appropriately selected comparison group(s), and should be analyzed by study arm and dose, as applicable.

3.3. Data presentation

These guidelines represent a desirable standard for the presentation and publication of data on encephalitis/ADEM to allow for comparability of data of data, and are recommended as an addition to data presented for the specific study question and setting. Additionally, it is recommended to refer to

existing general guidelines for the presentation and publication of randomized controlled trials, systematic reviews, and meta-analyses of observational studies in epidemiology (e.g., statements of Consolidated Standards of Reporting Trials [CONSORT], of Improving the quality of reports of meta-analyses of randomised controlled trials [QUORUM], and of Meta-analysis Of Observational Studies in Epidemiology [MOOSE]), respectively [65–67].

- (32) All reported events of encephalitis/ADEM should be presented in case-report fashion, where applicable. In the event that sufficient numbers of events/cases are available to present in tabular or collated format, this presentation could be used.
- (33) All reported cases of encephalitis/ADEM should include the calculation of the variables, preferably in table form, outlined in “data analysis” above.
- (34) Data on encephalitis/ADEM should be presented in accordance with data collection guidelines 1–21 and data analysis guidelines 25–31.
- (35) Data should be presented with numerator and denominator (n/N) and not only in percentages.

Although in immunization safety surveillance systems denominators are usually not readily available, attempts should be made to identify approximate denominators. The source of the denominator data should be reported and calculations of estimates described (e.g., obtained from manufacturer, Ministry of Health and coverage/population based data).

- (36) If the distribution of data is skewed, the median and range are usually the more appropriate statistical descriptor than a mean. However, the mean and standard deviation also should be provided.
- (37) The incidence²⁵ and prevalence²⁶ of cases in the study population should be presented and clearly identified as such in the text.
- (38) Any publication of data on encephalitis/ADEM should include a detailed description of the methods used for data collection and analysis. It is essential to specify
 - The study design of clinical trials;
 - For surveillance systems:
 - The type of surveillance system (e.g., passive surveillance, active surveillance);
 - The characteristics of the surveillance system (e.g., population served, mode of report solicitation);
 - The search strategy in surveillance databases;
 - Comparator group(s), if used for analysis;
 - Whether the day of immunization was considered “day 1” or “day 0” in the
 - analysis;

- Whether the date of onset²⁰ and/or the date of first observation²⁰ and/or the date of
- diagnosis²¹ was used for analysis;
- Reference of the case definition(s) used (Brighton or other) in the abstract or methods section of a publication.²⁷

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Appendix A

The validated tools in the appendix have been listed by this working group to assist the user of the proposed case definition in identifying possible tools based on their widespread use or experience with these tools in previous studies, and because use of identical tools in study settings would further allow for data comparability. Suggestion of these tools does not imply official endorsement of proprietary products by the Centers for Disease Control and Prevention.

A.1. Glasgow Coma Score, Pediatric Coma Scale [68,69]

Glasgow Coma Score

The GCS is scored between 3 and 15, 3 being the worst, and 15 the best. It is composed of three parameters: Best Eye

²⁶ For example, cases of encephalitis/ADEM on day 1: 2 cases, day 2: 10 cases, day 3: 3 cases, etc.

²⁷ Use of this document should preferably be referenced by referring to the respective link on the Brighton Collaboration website (<http://www.brightoncollaboration.org>).

Response, Best Verbal Response, Best Motor Response, as given below

Best Eye Response. (4)

1. No eye opening
2. Eye opening to pain
3. Eye opening to verbal command
4. Eyes open spontaneously

Best Verbal Response. (5)

1. No verbal response
2. Incomprehensible sounds
3. Inappropriate words
4. Confused
5. Orientated

Best Motor Response. (6)

1. No motor response
2. Extension to pain
3. Flexion to pain
4. Withdrawal from pain
5. Localising pain
6. Obeys commands

It is important to break the figure down into its components, such as E3V3M5 = GCS 11.

A Coma Score of 13 or higher correlates with a mild brain injury, 9–12 is a moderate injury and 8 or less a severe brain injury.

Pediatric Coma Scale

Eyes open	Best verbal response	Best motor response
☐ Spontaneously (4)	☐ Orientated (5)	☐ Obeys command (5)
☐ To speech (3)	☐ Words (4)	☐ Localizes pain (4)
☐ To pain (2)	☐ Vocal sounds (3)	☐ Flexion to pain (3)
☐ None (1)	☐ Cries (2)	☐ Extension to pain (2)
	☐ None (1)	☐ None (1)

Total =

Adjustment to age

- During the first 6 months
 - The best verbal response is normally a cry, though some infants make vocal responses during this period. Normal verbal score expected is 2.
 - The best motor response is usually flexion. Normal motor score expected is 3.
- 6–12 months
 - The normal infant makes noises: normal verbal score expected is 3.
 - The infant will usually locate pain but not obey commands: normal motor score expected is 4.
- 12 months to 2 years
 - Recognizable words are expected: normal verbal score expected is 4.
 - The infant will usually locate pain but not obey commands: normal motor score expected is 4.

- 2–5 years
 - Recognizable words are expected: normal verbal score expected is 4.
 - The infant will usually obey commands: normal motor score expected is 5.
- After 5 years
 - Orientation is defined as awareness of being in hospital: normal verbal score expected is 5.

Normal aggregate score

0–6 months	9
>6–12 months	11
>1–2 years	12
>2–5 years	13
>5 years	14

A.2. Features that may aid in distinguishing acute disseminated encephalomyelitis (ADEM) from Multiple Sclerosis—per suggestions by this Working Group

	ADEM	Multiple Sclerosis
Prodromal febrile illness	Common	Unusual
Signs and symptoms	Frequent widespread CNS disturbance; coma/drowsiness common	Frequently monosymptomatic
Temporal pattern of illness	Monophasic	Relapsing and remitting
Neuroimaging features	High lesion load Large, bilateral white matter lesions Thalamic involvement sometimes present Lesions of same age	Lower lesion load (generally) Smaller plaques in deep white matter Thalamic or other deep grey involvement unusual Lesions of different ages
Cerebrospinal fluid	Oligoclonal bands frequently absent	Oligoclonal bands frequently present

A.3. Mini-Mental State Examination [70]

A.3.1. Questions (Total of 30 points)

- A. Orientation (10 points)
 1. Year, season, date, day of week, and month
 2. State, county, town or city
 3. Hospital or clinic, floor
- B. Registration (3 points)
 1. Name three objects: apple, table, penny
 2. Each one spoken distinctly and with brief pause
 3. Patient repeats all three (one point for each)
 4. Repeat process until all three objects learned
 5. Record number of trials needed to learn all three objects

- C. Attention and calculation (5 points)
1. Spell WORLD backwards: DLROW
 2. Points given up to first misplaced letter
 3. Example: DLORW scored as 2 points only
- D. Recall (3 points)
1. Recite the 3 objects memorized in Registration above
- E. Language (9 points)
1. Patient names two objects when they are displayed
 - a. Example: pencil and watch (1 point each)
 2. Repeat a sentence: 'No ifs ands or buts'
 3. Follow three stage command
 - a. Take a paper in your right hand
 - b. Fold it in half
 - c. Put it on the floor
 4. Read and obey the following
 - a. Close your eyes
 - b. Write a sentence
 - c. Copy the design (picture of 2 overlapped pentagons)

A.3.2. Interpretation of Mini-Mental State Score (maximum: 30)

- A. Normal score: 24 or higher
- B. Educational and age norms
1. Fourth grade education
 - a. Ages 18–69: median MMSE score 22–25
 - b. Ages 70–79: median MMSE score 21–22
 - c. Age over 79: median MMSE score 19–20
 2. Eighth grade education
 - a. Ages 18–69: median MMSE score 26–27
 - b. Ages 70–79: median MMSE score 25
 - c. Age over 79: median MMSE score 23–25
 3. High school education
 - a. Ages 18–69: median MMSE score 28–29
 - b. Ages 70–79: median MMSE score 27
 - c. Age over 79: median MMSE score 25–26
 4. College education
 - a. Ages 18–69: median MMSE score 29
 - b. Ages 70–79: median MMSE score 28
 - c. Age over 79: median MMSE score 27

A.4. Barthel index [71]

**Activity
Score**

Feeding

0 = unable
5 = needs help cutting, spreading butter, etc.,
or requires modified diet
10 = independent

0 5 10

Bathing

0 = dependent
5 = independent (or in shower)

0 5

Grooming

0 = needs to help with personal care
5 = independent face/hair/teeth/shaving (implements provided)

0 5

Dressing

0 = dependent
5 = needs help but can do about half unaided
10 = independent (including buttons, zips, laces, etc.)

0 5 10

Bowels

0 = incontinent (or needs to be given enemas)
5 = occasional accident
10 = continent

0 5 10

Bladder

0 = incontinent, or catheterized and unable to manage alone
5 = occasional accident
10 = continent

0 5 10

Toilet Use

0 = dependent
 5 = needs some help, but can do something alone
 10 = independent (on and off, dressing, wiping)
 0 5 10

Transfers (bed to chair and back)

0 = unable, no sitting balance
 5 = major help (one or two people, physical), can sit
 10 = minor help (verbal or physical)
 15 = independent
 0 5 10 15

Mobility (on level surfaces)

0 = immobile or < 50 yards
 5 = wheelchair independent, including corners, > 50 yards
 10 = walks with help of one person (verbal or physical) > 50 yards
 15 = independent (but may use any aid; for example, stick) > 50 yards
 0 5 10 15

Stairs

0 = unable
 5 = needs help (verbal, physical, carrying aid)
 10 = independent
 0 5 10

TOTAL (0 - 100)

Patient Name: _____ Rater: _____ Date: ____/____/____:____

THE BARTHEL ADL INDEX: GUIDELINES

1. The index should be used as a record of what a patient does, not as a record of what a patient could do.
2. The main aim is to establish degree of independence from any help, physical or verbal, however minor and for whatever reason.
3. The need for supervision renders the patient not independent.
4. A patient's performance should be established using the best available evidence. Asking the patient, friends/relatives and nurses are the usual sources, but direct observation and common sense are also important. However direct testing is not needed.
5. Usually the patient's performance over the preceding 24-48 hours is important, but occasionally longer periods will be relevant.
6. Middle categories imply that the patient supplies over 50 per cent of the effort.
7. Use of aids to be independent is allowed.

A.5. Modified Rankin Scale (MRS) [72]

Score	Description
0	No symptoms at all
1	No significant disability despite symptoms; able to carry out all usual duties and activities
2	Slight disability; unable to carry out all previous activities, but able to look after own affairs without assistance
3	Moderate disability; requiring some help, but able to walk without assistance
4	Moderately severe disability; unable to walk without assistance and unable to attend to own bodily needs without assistance
5	Severe disability; bedridden, incontinent and requiring constant nursing care and attention
6	Dead
SCORE (0-6): _____	
Patient Name: _____	
Rater: _____	Date: ____/____/____ ____:____

A.6. Selected testing for infectious etiologies of acute encephalitis (note: not all testing will be available at all institutions)—per suggestions by this Working Group

Causative agent	Specimen	Test(s)	Abbreviations
		A/C serum	Acute and convalescent sera
		CSF	Cerebrospinal fluid
		NP swab	Nasopharyngeal swab
		Wh blood	Whole blood
		PCR	Polymerase chain reaction
		IFA	Immunofluorescence assay
Tier I agents: common etiologies of acute or post-infectious encephalomyelitis worldwide			
<i>Bartonella hensellae</i>	CSF	PCR, serology, culture	“Cat-scratch disease”
	A/C serum	PCR, serology, culture	
CMV	CSF	PCR, serology, culture	Cytomegalovirus
	A/C serum	PCR, serology, culture	
<i>Cryptococcus neoformans</i>	CSF	PCR, antigen detection, identification of yeast on India ink staining, culture	More common in immunosuppressed/HIV+
	A/C serum	serology	
EBV	CSF	PCR, serology, culture	Epstein-Barr virus
	A/C serum	PCR, serology, culture	
Enteroviruses (various strains)	Stool/rectal swab	Culture	
	CSF	PCR, serology, culture	
HHV-6	CSF	PCR, serology, culture	Human herpes virus 6
	A/C serum	PCR, serology, culture	
HIV	Serum	Rapid antibody testing, PCR	Human immunodeficiency virus
	CSF	PCR	

Appendix A.6 (Continued)

HSV1	CSF	PCR, serology, culture	Herpes simplex virus 1
	A/C serum	PCR, serology, culture	
HSV2	CSF	PCR, serology, culture	Herpes simplex virus 2
	A/C serum	PCR, serology, culture	
JC virus	CSF	PCR	Progressive multifocal leukoencephalopathy; more common in immunosuppressed/HIV +
<i>Mycobacterium tuberculosis</i>	CSF	Identification of acid-fast bacilli; culture	
	Serum	Identification of acid-fast bacilli; culture	
Rabies virus	CSF/brain biopsy	PCR, IFA, culture, serology	
	Saliva	PCR, IFA, culture	
	Nuchal skin biopsy	PCR, IFA	
<i>Treponema pallidum</i>	CSF	Rapid plasminogen reaction (RPR), VDRL, immunohisto-chemistry, culture; followed by confirmation by tests utilizing treponemal antigens (FTA-Abs, MHA-TP, TPHA)*; immunohisto-chemistry, culture	Neurosyphilis
FTA-Abs = fluorescent treponemal antibody adsorbed			
MHA-TP = microhemagglutination assay for treponema pallidum			
TPHA = <i>T. pallidum</i> hemagglutinating antibody			
	A/C serum	RPR, VDRL; followed by treponemal antigen testing; immunohisto-chemistry, culture	
<i>Tropheryma whippelli</i>	CSF	PCR, serology	Whipples's disease
	A/C serum	PCR, serology	
	Intestinal or brain tissue	PCR	
VZV	CSF	PCR, serology, culture	Varicella-zoster virus
	A/C serum	PCR, serology, culture	
Tier 2: common etiologies of acute encephalomyelitis, geographically restricted			
Arboviruses			
<i>Alphaviridae</i> ^a	CSF	PCR, serology	
	A/C serum	PCR, serology	
<i>Bunyaviridae</i> ^b	CSF	PCR, serology	
	A/C serum	PCR, serology	
<i>Flaviviridae</i> ^c	CSF	PCR, serology	
	A/C serum	PCR, serology	
<i>Borrelia burgdorferi</i>	CSF	PCR, serology	Lyme disease
	A/C serum		
<i>Leptospira</i> spp.	CSF	PCR, serology, microagglutination testing	
	A/C serum	PCR, serology, microagglutination testing	
<i>Plasmodium</i> spp. (Malaria)	Wh blood	Blood smear	

Appendix A.6 (Continued)

Tier 3: Less common etiologies, infrequently associated with acute encephalomyelitis

Adenovirus	NP/throat swab	PCR, culture	
<i>Angiostrongylus cantonensis</i>	CSF, brain tissue	Parasite isolation (presence of eosinophilia in CSF is suggestive)	Eosinophilic meningitis
	A/C serum	Serology	
<i>Chlamydia pneumoniae</i>	A/C serum	Serology, PCR	
<i>Ehrlichia chaffeensis</i>	A/C Serum	Identification of antigen by IFA, serology, PCR, culture	
	CSF	PCR, serology, culture	
	Wh blood	PCR (detection of nucleic acid in leukocytes)	
<i>Entamoeba histolytica</i>	CSF, brain tissue	Microscopic identification of protozoa	Amoebiasis
	Fecal specimens, smears of aspirates	Microscopic identification of protozoa	
<i>Gnathostoma spinigerum</i>	CSF	Parasite isolation	Gnathostomiasis
	Skin biopsy	Parasite isolation	
Influenza A&B viruses	NP/throat swab	Antigen detection, PCR, culture	
	A/C serum	Serology	
Measles virus	CSF	PCR, serology, culture	
	A/C serum	PCR, serology, culture	
Mumps virus	CSF	PCR, serology, culture	
	A/C serum	PCR, serology, culture	
<i>Mycoplasma pneumoniae</i>	A/C serum	Serology, PCR, culture	
	CSF	Serology, PCR, culture	
Parainfluenza virus	NP/throat swab	Antigen detection, PCR, culture	
	A/C serum	Serology, PCR	
Parvovirus B-19	A/C serum	Identification of motile amoeba in fresh CSF, or identification in stained CSF mounts; culture	Includes agents <i>Naegleria fowleri</i> , several species of <i>Acanthamoeba</i> , and <i>Balamuthia mandrillaris</i>
Primary amoebic meningoencephalitis	CSF		
<i>Rickettsia rickettsii</i> (RMSF)	A/C serum	Serology, antigen by IF	
	Skin biopsy	Antigen detection by IF	

[§]The list of specimens to be collected, and the diagnostic assays, has not necessarily been placed in order of diagnostic preference or utility. Different methods of laboratory diagnostic testing will undoubtedly differ in various settings. In general, testing involving CSF or brain tissue is the most convincing way of demonstrating an infectious central nervous system process; however, tests involving CSF/brain tissue are not always the most sensitive (or specific) for various agents. [¶]This list has purposely excluded agents causing primarily bacterial meningitis (*Neisseria meningitidis*, *Streptococcus pneumoniae*, etc.). While these agents can present clinically as acute encephalitis, it is presumed that such agents will be tested for in nearly all situations of apparent CNS infection. Similarly, this list has purposely excluded most fungal CNS infections, as fungal CNS disease tends to have a more prolonged, chronic course and generally differs clinically from acute encephalitis. If such agents are suggested by clinical findings and epidemiology, they should be excluded as well. [~]The table has attempted to include the most common among agents leading to acute or post-infectious encephalomyelitis, to aid in clinical test decision-making. It is not intended to be an exhaustive list, and additional agents to be excluded may be suggested by clinical features, geography, or epidemiology.

^a Includes Eastern equine encephalitis virus (EEEV), Western equine encephalitis virus (WEEV), Venezuelan encephalitis virus (VEEV).

^b Includes LaCrosse virus (LACV), Jamestown Canyon virus (JCV), Snowshoe hare virus, Cache Valley virus, other California serogroup viruses.

^c Includes St. Louis encephalitis virus (SLEV), West Nile virus (WNV), Japanese encephalitis virus (JEV), Tick-Borne encephalitis virus (TBEV), Dengue virus (DENV), Murray Valley encephalitis virus (MVEV), Rocio virus, other viruses infrequently associated with human illness.

A.7. Additional testing in acute encephalitis/ADEM
(note: not all testing will be available at all institutions)—per suggestions by this Working Group

Tests

Serum

Complete blood count (CBC)
Leukocyte differential
Erythrocyte sedimentation rate (ESR)
Alanine aminotransferase (ALT)
Total protein
Alkaline phosphatase
Electrolytes
Calcium
Glucose
Thyroid-stimulating hormone
Vitamin B12 level
Folate level
Toxicology screen
Serum protein electrophoresis (SPEP)
Anti-nuclear antibody (ANA)
Rheumatoid factor (RF)
Anti-double stranded DNA antibody
SS-A (Ro), SS-B (La)
Anti-cardiolipin antibody
Angiotensin-converting enzyme
Lupus anticoagulant

CSF

IgG index
IgG synthesis rate
Oligoclonal bands
Myelin basic protein
Venereal disease research laboratory (VDRL), rapid plasma regain (RPR)

Culture (specimens are nasopharyngeal or rectal swabs)

Plate-count monensin KCl (PMK) culture media
MRC-5 human diploid cells
A-549 lung carcinoma cell line
Rotaviral antigen testing
Human fetal diploid kidney (HFDK) cells
Human fetal diploid lung (HFDL)

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